

Speedy Prints

Product:

The product of this project is a fully working **3D printer**. This would be mostly inspired by the Prusa i3 MK3S+, a very capable printer with a relatively easy design, with the only difference being that my printer would be bigger in the X and Y axis since my prints are usually big but not very tall. The goal would be a 30x30x30cm size 3D printer(though as you may see in the reflection, it didn't work out).

Learning Goal:

The goal would be to learn how to take my knowledge about programming and electronics and expand it to new levels. This would be achieved by designing and/or assembling each and every part of my 3D printer, from the structure to the electronics.

Personal Interest:

I really like programming. I have made various websites, like a science calculator website. But as well as programming, I also like designing and printing different parts to fix things, create cool tools or just create interesting things. But I had one issue, my parents were never going to buy me a 3D Printer, they were simply too expensive. And thus, I decided to make it myself and use personal project as a valid reason for them to fund it.

And thus it started:

Planning:

The project would be divided into various parts each taking as little as 1-2 weeks to as much as a couple months:

1. Planning (1 week)
2. Research (1 month)
3. Budgeting (2 weeks)
4. Design parts that couldn't be bought or just didn't make sense to buy (1.5 months)
5. Assemble and modify parts that didn't fit correctly (2.5 months)
6. Programm and calibrate (1 week)
7. Present (1 day)

And though this seems pretty simple(I thought the same) these steps would be incredibly hard, especially steps 3 and 4 which took over 4ish months to complete since each part would usually take a minimum of 2-3 days and in some cases even 1-2 weeks to design and print to a successful degree.

Success Criteria:

1. Build is stable and sturdy
2. Build is done in time
3. 3D printer has decent print quality
4. 3D printer can print in PLA without heated bed

5. Electronics is not a wire mess

Applying Skills

Applying skills for Learning Goal:

Research skills:

These skills were used to find out how 3D printers work, what parts I should use for mine, and what firmware I should use.

Before getting into all the complicated parts of the printer, I had to learn the basics. Like how stepper motors work and how to drive them with an arduino. This is because stepper motors are, at least I would say, the most important part of the 3D printer; without them the printer wouldn't work.

There are 2 types of stepper motors (that I know of) which are Bipolar and Unipolar. Bipolar works by switching the polarity of 2 coils, each placed all around the rotor (the shaft of the motor) this creates a very precise controlled movement that can be used for 3D printers, they are called Bipolar because the current can run both ways inside the motor. On the other hand, Unipolar stepper motors use more coils and have 1-2 extra wires since they have extra VM wires as seen in figure 1. They are called Unipolar because current can only flow one way through them.

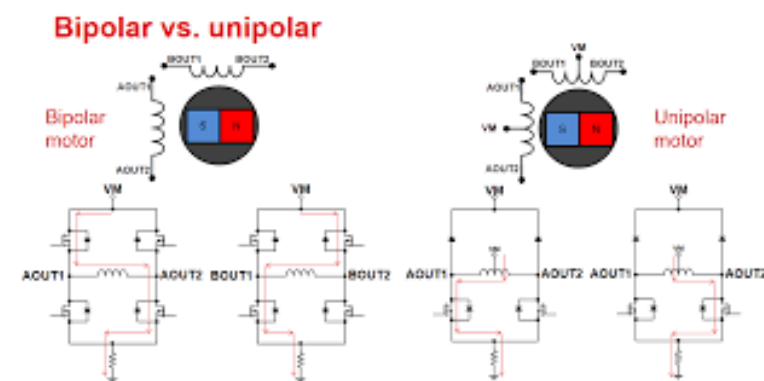


Fig 1: Schweber, Bill. (6/1/2022) Unipolar vs bipolar circuit diagram [photo] Power Electronics Tips. <https://www.powerelectronicstips.com/unipolar-vs-bipolar-drive-for-stepper-motors-part-1-principles-faq/>

How to drive the stepper motors:

In this case I will be using an arduino UNO rev 3 and a L239D H-Bridge to run the motor. The wiring setup is the one shown in figure 2. This accompanied by some code, was able to drive the motor and teach me the basics of how to run a stepper motor.

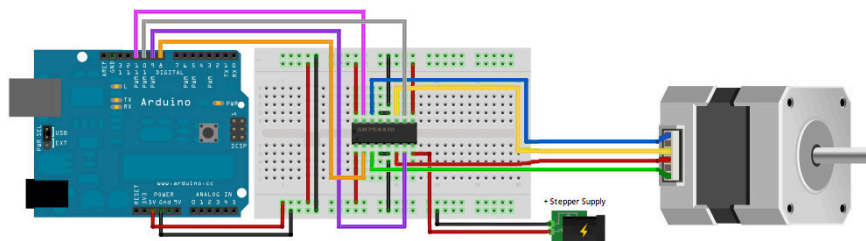


Fig 2: Forman, Eric. (31/3/2013) how to run stepper motor with arduino [photo]. Hello Circuits. <https://hellocircuits.com/2013/03/31/stepper-motor/>

Best extruder/Hotend:

The extruder and Hotend are the bits of the printer that interact with the plastic, The extruder pushes the plastic and the hot end melts it to come out of a 0.4 mm hole. The options for the extruder/hotend assembly were: the E3D Hemera, the BIQU H2 V2S, Bondtech BMG, Micro Swiss Direct Drive Extruder, and the E3D Titan extruder/V6 hot end. The requirements for the parts were the following: good extrusion quality, Compatibility with Marlin 2 and the ramps 1.4 3D, and reasonable price.

E3D Hemera:

The E3D Hemera is a great Extruder, one of the best in the market. It has many features like the specially designed heat sink which means that the cooling of the hot end is custom made to increase the extruder's efficiency to the max. As well as that the Hemera also has Dual drive extrusion, which basically means that the filament is pushed with gears of either side of it instead of only one side of it, unlike most printers.

Nevertheless, the price can't be ignored. The E3D Hemera is a very expensive extruder, coming in at over 120 USD, sometimes even over 200 USD. This price means that this is not the extruder for this project.

BIQU H2 V2S:

The BIQU H2 V2S is a very capable extruder, with very similar prospects to the Hemera, with a very good print quality and compatibility with Marlin 2 and the Ramps 1.4. All of that at a lower price as well.

And though at the beginning of the research stage I mostly found that it was made for Ender printers, I have now done some more thorough research and found that the BIQU would have also been a great choice for the project, maybe even a better one due to the full integration of the extruder and Hotend. To this I can also add ease of assembly, Quality components Upgradability and Community Support since this is a common choice for DIY printers. The only issue would be scarcity of parts for this particular extruder where I live.



Fig 3: Biqu. (n.d.). Biqu H2 VS2 [photo]. Biqu.
<https://biqu.equipment/collections/h2-series-extruder/products/biqu-h2-v2s-extruder-for-b1-bx-ender-3-3-v2-5-6-cr6-10>

BondTech BMG:

The Bondtech BMG is an Accurate and precise 3D printer extruder. It has the dual extruder that the Hemera has, which allows for precise filament control, better retraction performance, reduced filament slippage and enhanced print speeds. As Well as this, the BMG is also compatible with Marlin 2 and the Ramps 1.4.

All of this is great, but the Bondtech BMG does come at a price of around 80-90 dollars for a genuine part just for the extruder (without the hot end). This means that this is way out of my budget.

Micro Swiss Direct Drive Extruders:

These are all Extruders made by an American company that makes mostly all metal extruders which means their extruders have some of the same qualities as the rest in this list with good print quality and compatibility with Marlin 2 and the Ramps 1.4 3D. Added to this, they also have a more compact design which means that the extruder and hotend are closer together leading to better retraction, and reduced risk of artifacts like blobs and stringing.

Though that is all great, the all metal construction does come at a premium meaning their extruders are out of my budget.

E3D Titan and V6 Hot End:

The E3D Titan extruder is relatively lightweight, comes in a compact shape, has a vast amount of replacement parts, and supports various filament sizes. It also fits in my budget at 44.55 (this may change since it is transformed from GBP) US dollars. Other than that, it is also modular which means I can upgrade it down the road, change broken parts and mount it to the 3D printer easier.

As well as that the V6 Hot end is also very easy to assemble and replace parts, has a great community for support and can be natively assembled with the E3D titan. It also has great compatibility with Marlin 2 and Ramps 1.4 which makes it easier to integrate into the 3D printer. The V6 also has a great price at around 65 dollars.

The Titan and V6 were the extruder/hotend used for the project, though upon further research the Biqu H2 V2S seems to be a better option.

Best 3D printer Layout/Design

In this case, due to the simplicity of the design, I will use a very similar design to the one of the Prusa i3 MK3S+ that we have at school. This design mostly consists of 2 main aspects: Frame and structure, and Motion systems. The frame and Structure is mostly made of aluminum extrusions and injection molded plastic parts, which make the structure of the printer very stable and sturdy. Then, the linear motion systems allow for precise movement with the use of linear bearings, belts and screw shafts to reduce friction and improve accuracy.



Fig 4: Prusa. (n.d.). Original Prusa i3 MK3S+ 3D printer [photo]. Prusa Research. <https://www.prusa3d.com/product/original-prusa-i3-mk3s-3d-printer-3/>
Picture of a Prusa i3 MK3S+

Is the bed heater required?

According to my research: Yes and no, it depends on the filament that you are planning to use. For example: PLA doesn't require a heated bed, but does prefer one, as said in this article: "printed at a low temperature and does not require a heated bed" (Simplify3D Software, 2019). Although other materials like ABS do need a heated bed, in the case of ABS the bed must be from 95° to 110° Celsius.

Tests:

Many test were done to determine the need of a heated bed but the documentation of the tests wasn't done properly by myself. Thus I only have image as evidence of the test:

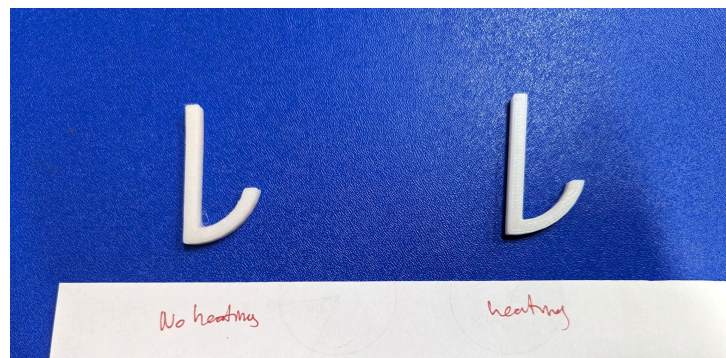


Fig 5: Riboldi, Matias. (19/10/2023). heated bed test [photo].

These were pieces meant to be able to test the material adhesion in layers and also the bed adhesion. Some of them were printed with supports as well (just to see what happens). During

the tests none of the prints failed due to the part being detached from the bed even when using supports. Thus I would consider that PLA can be printed without a heated bed though I will add glue to my printer's bed just to ensure the part sticks to the bed.

Best Microcontroller

There are many microcontrollers out there, but I decided to use either the Raspberry pi 4B or the Arduino MEGA 2560 due to availability, community support and ease of use. Before I bought them I decided to test them just to be sure I bought the right thing.

The raspberry pi 4B seemed to be the right choice since it was very capable, and had many ports to be able to pass the files through ethernet or USB. But During my test the Raspberry Pi didn't even boot up. There seemed to be an issue with the SD card, which had to be changed, then the Raspberry pi seemed to get stuck in a cleaning up loop which didn't stop no matter how hard I tried. Thus the Raspberry Pi was out as an option.

The Arduino MEGA 2560, is a very popular arduino board that can be used for everything from rocket models to 3D printers. This is programmed with the normal arduino IDE and, conveniently, was compatible with a board that is commonly used to control and wire up the entire 3D printer. Due to all of this, the Arduino MEGA was the Microcontroller of choice.

Self-Management skills:

Budgeting:

In the case of this project, the budget would be under 900 USD, since that is what a fully assembled Prusa i3 MK3S+ costs. Why compare my 3D printer to the Prusa? Well, it is mostly because that is the printer that we have at school, thus I am able to compare the results to a much more accurate degree. In the table below I have displayed all the prices, quantities, specific part names, manufacturers of each part and a link to the site in which they are sold.

a.

Part Name	Name	Link	Price	Quantity
Extruder	E3D Titan	https://e3d-online.com/products/titan-extruder	\$32.73	1
Hot End	E3D V6	https://e3d-online.com/products/v6-all-metal-hotend	\$67.68	1
Stepper Motor	Nema (XY42STH34-0354A) 17	https://www.crcibernetica.com/stepper-motor-nema-17-xy42sth34-0354a/	\$21.95	5
Microcontroller	Arduino MEGA	https://www.amazon.com/ARDUINO-MEGA-2560-REV3-A000067/dp/B0046AMGW0?crid=3K35764O5RTOX&keywords=Arduino+Mega+2560+rev+3&qid=1707240792&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&srefix=arduino+mega+2560+rev+3.aps.389&sr=8-1-spons	\$48.90	1
2020 Extrusion	2020 V-Slot Aluminum	https://www.crcibernetica.com/2020-v-slot	\$6.95	5

	Extrusion (500mm)	-aluminum-extrusion-500mm/		
Belts	10mm Black GT2 Belt (1 meter)	https://www.crcibernetica.com/10mm-black-gt2-belt-1-meter/	\$7.95 per meter	3 meters
Non-heated glass bed	Tempered glass bed	N/A	free	1
Screws	M3 screws in different sizes	https://www.crcibernetica.com/m3-stainless-Phillips-steel-screw-and-nut-assorted-kit-240pcs/	\$18.95	1
Special nut for 2020 V-slot extrusion	M3 T-nut for 2020 V-slot extrusion	https://www.crcibernetica.com/10pcs-m3-t-nut-for-2020-aluminum-extrusion/	\$1.25	2
8mm 50cm rod	8mm 50cm smooth rod	https://www.crcibernetica.com/8mm-smooth-rod-50-cm-long/	\$4.95	5
Linear Rail Shaft Guide/Support (8mm)	Linear Rail Shaft Guide/Support (8mm)	https://www.crcibernetica.com/linear-rail-shaft-guide-support-8mm-diameter-sk8/	\$2.95	6
Linear Bearing (8mm)	Linear Bearing Platform (8mm)	https://www.crcibernetica.com/linear-bearing-platform-8mm-diameter-sc8uu/	\$4.95	4
Screw shaft	T8x500mm Lead Screw with Brass Nut	https://www.crcibernetica.com/t8x500mm-lead-screw-with-brass-nut/	\$10.95	2
5mm to 8mm shaft coupler	Flexible Coupler 5x8x25mm	https://www.crcibernetica.com/flexible-coupler-5x8x25mm/	\$2.95	2
Belt to shaft adapter	GT2 DRIVE Pulley 20T 10mm (5mm bore)	https://www.crcibernetica.com/gt2-drive-pulley-20t-10mm-5mm-bore/	\$2.95	2
Bearing	608ZZ Roller Bearing	https://www.crcibernetica.com/608zz-roller-bearing/	\$2.95	3
PSU	12V 10A power supply	https://www.crcibernetica.com/12v-10a-power-supply/	\$16.95	2 (one extra just in case)
		Total:	552.01 USD	100 extra (for shipping costs)

Applying skills for product:

Thinking skills, Mainly Problem Solving and Troubleshooting:

This step involves lots of troubleshooting and Problem solving. This is because most of the parts were designed by me and often didn't fit right the first time. Thus, I had to find out what was wrong, find a way to fix it and then implement it. For all the parts.

Building the 3D printer:

This was the longest step taking over 3ish months. It was all done as I went.

I started with a general assembly of the base as seen in figure 6 and 7:

In figure 6 and 7 the black parts are 2020 Extrusion, the long metal parts are 8mm rods, the metal bits at the end of the rods and the rod holders, and the rectangular metal bits that are on the rod are linear bearings to guide the movement of the bed. Joining the black metal parts are some small 90° 2020 extrusion couplers with screw holes to make sure that they don't move. After I had this, I proceeded to design the Y-axis motor mount.

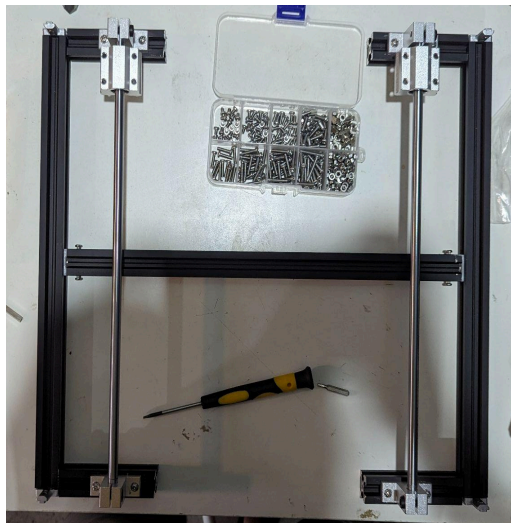


Fig 6: Riboldi, Matias. (22/1/2024). fig 6 [photo].

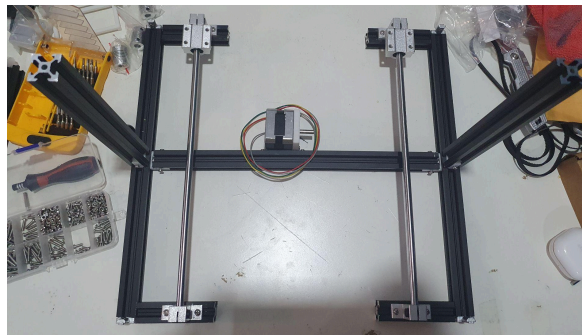


Fig 7: Riboldi, Matias. (25/1/2024). fig 7 [photo].

After this, I decided to start prototyping the joint between the 2 linear bearings, and added the Z-axis motor mounts. These mounts are made of 3D printed 2020 extrusion couplers and some laser cut wooden parts that lock into the 3D printed parts with a tight fit and hold the Z-axis motors. These motors have a 5mm to 8mm shaft adapter that joins the motor shaft and a 500mm screw shaft.

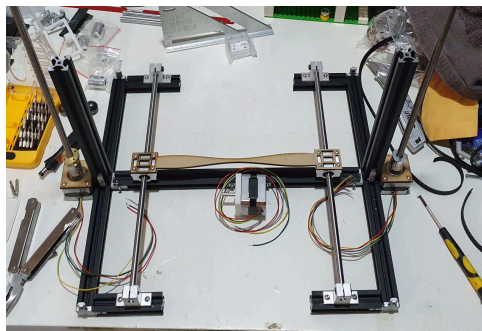


Fig 8: Riboldi, Matias. (29/1/2024). fig 8 [photo].

In figure 9 you can see a test of the motors. This was a test to make sure that the motors worked. It was performed with an arduino UNO Rev 3, using a 9V battery and 3 L239D H-Bridges. [OBJ]

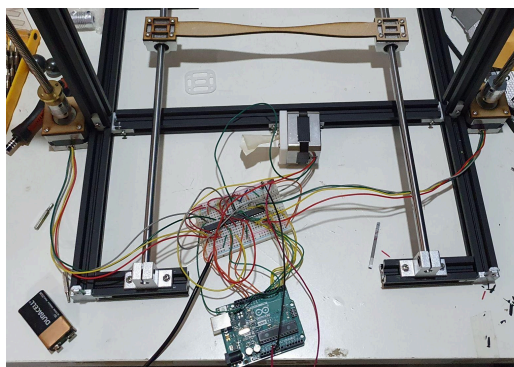


Fig 9: Riboldi, Matias. (31/1/2024). fig 9 [photo].

Shortly after these tests, another test was performed to determine the capabilities of the Raspberry Pi 4B, which as you probably read previously, didn't workout.



Fig 10: Riboldi, Matias. (5/2/2024). fig 10 [photo].

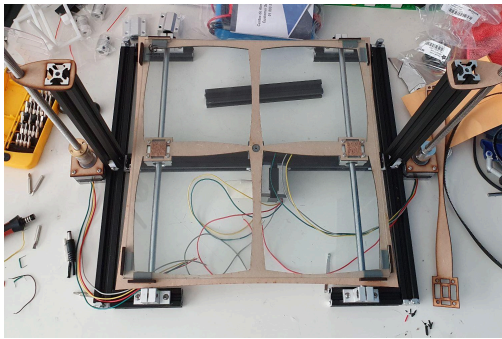


Fig 11: Riboldi, Matias. (6/2/2024). fig 11 [photo].

Then, in figure 11, you can see that I added the complete version of the joint between the 2 linear bearings and then a tempered glass on top of that which will act as the print bed. As well as that, I also added, what I call Z top mounts which are the wooden pieces on top of the Z-axis 2020 extrusion.

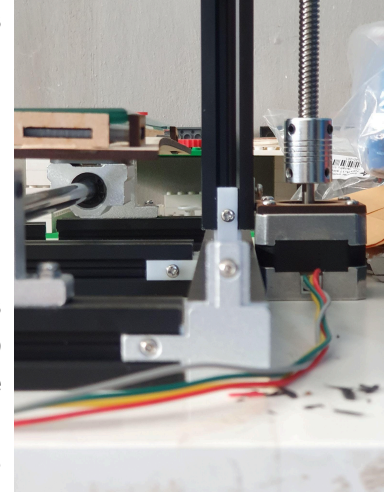


Fig 12: Riboldi, Matias. (8/2/2024). fig 12 [photo].

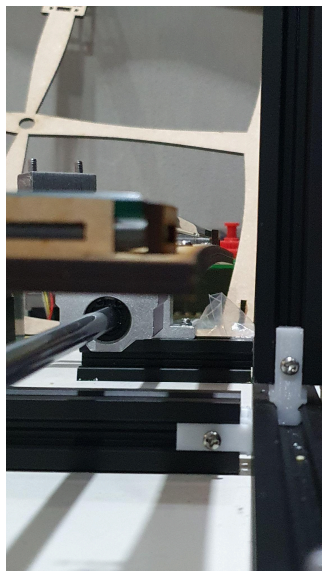


Fig 13: Riboldi, Matias. (14/2/2024). fig 13 [photo].

In figure 12, you can see that the wooden plate that holds the glass is extremely bent, because of this, I decided to double the plates which sort of fixes the issue as seen in figure 13. Then, in figure 14 you can see how it looked once the double plates were in.

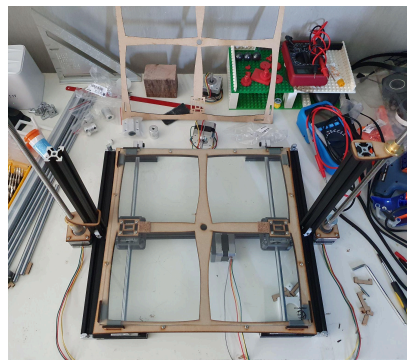


Fig 14: Riboldi, Matias. (15/2/2024). fig 14 [photo].

After this, in figure 15, I designed the first version of the X-axis mount to the Z-axis mount. This was done using special rollers for 2020 extrusion, and 8mm rods. This will hold the X-axis motor and belts, as well as the extruder/hot end.

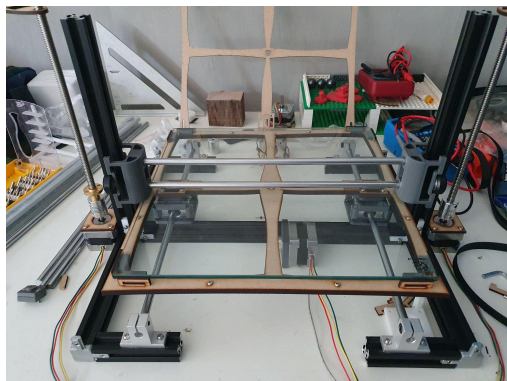


Fig 15: Riboldi, Matias. (16/2/2024). fig 15 [photo].

After this, on February the 20th I performed the first test of the motors with the new Arduino and Ramps boards along with a 12V power supply. This can be seen in figure 16.

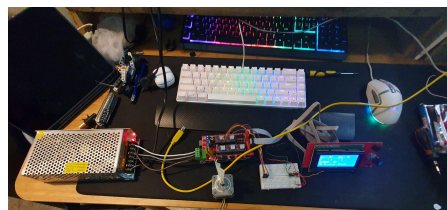


Fig 16: Riboldi, Matias. (20/2/2024). fig 16 [photo].

This test resulted in the first motor movement.

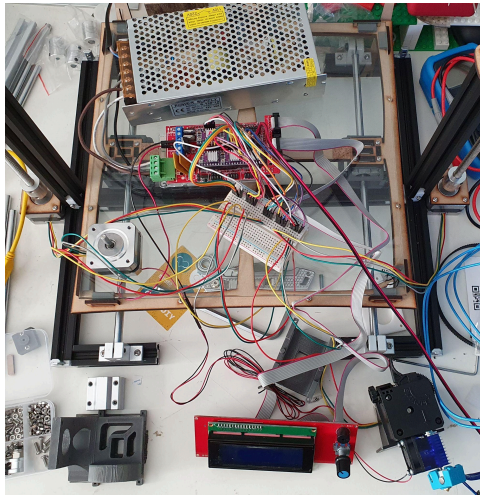


Fig 17: Riboldi Matias (21/2/2024). fig 17 [photo].

In figure 17 you can see the first test of all motors. This test was performed inside the 3D printer since some of the motors were already attached to the frame of the printer and would just take too much time to remove. Thankfully, the test was successful, resulting in correct motion in all the motors.

After this, I started designing and prototyping the X-axis motor mount. This process took a long time since the part had many complex requirements. As shown in figure 18, the first prototype didn't work since the distance between the screw shaft and the 2020 extrusion wasn't correct.

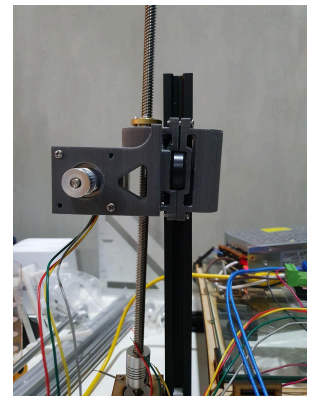


Fig 18: Riboldi, Matias (21/2/20224). [photo].

Once the issues with the X-axis motor mount were resolved, in figure 19, I performed a test of all the motors and the extruder/hotend assembly. Notable for this test is the fact that all the motors were placed in their final position for added accuracy of the results. (no belts were used in the test)

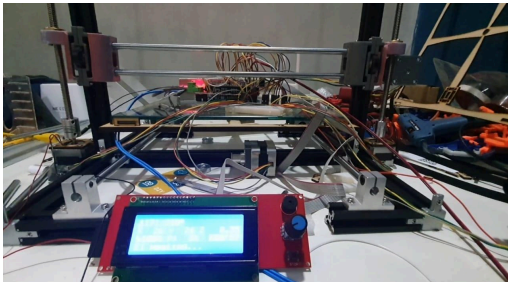


Fig 19: Riboldi, Matias. (22/2/2024). [photo].

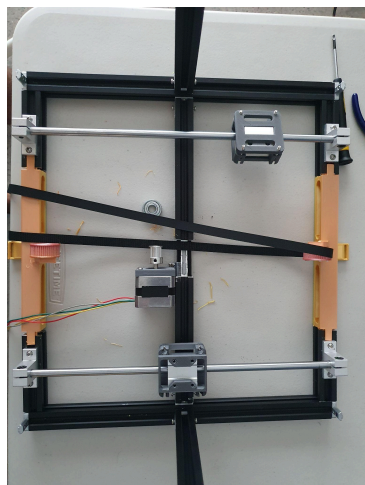


Fig 20: Riboldi, Matias. (27/2/2024). [photo].

In figure 20, there are some orange things that work as a joint between the two sides of the printer as well as a point to mount the belts on either side of the printer.

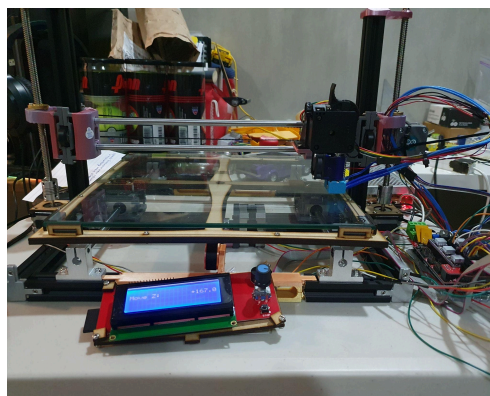


Fig 21: Riboldi, Matias. (28/2/2024). [photo].

In figure 21, you can see that the belts are now installed. Making the 3D printer almost capable of printing only missing calibration.

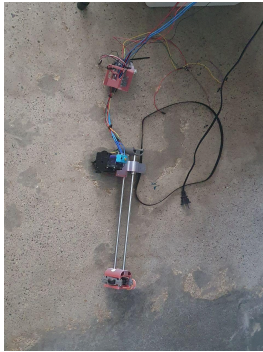


Fig 22: Riboldi, Matias.
(29/2/2024). [photo].

Then, disaster struck, the Thursday before presentation day(which was on a monday), all the X-axis assembly fell to the floor due to bad placement by me. This was a big error since re printing the parts would take over 15 hours. This was saved by an event that the school had on saturday which allowed me to start the print on friday and then pick it up on saturday. But, let's just say that I was really mad after the single most complicated and important part of the printer had just fallen so close to presentation day. But nonetheless, I carried on and mostly finished the project. (this will be talked about more during the reflection)

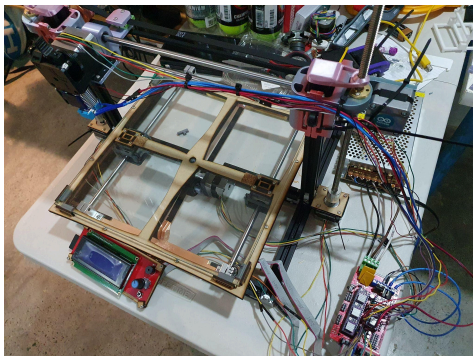


Fig 23: Riboldi, Matias. (2/3/2024).
[photo].

This is as close to completion as the printer got to. This was mostly due to extruder/hotend issues which prevented the molten plastic from exiting the nozzle. Everything else worked, so you could say that all the parts that I designed worked but not the ones that others designed(except the motor, extrusion and screw shaft).

Reflection

First of all, I'd like to note that the 3D printer never got to work during the span of the project. I will keep working on it till it works, but for now it doesn't. The reasons are the following:

1. **The Extruder didn't work.** During the testing of the 3D printer, the sunday before the exhibition, I got the nozzle and heat chamber clogged with filament. This was due to improper use of the fan which, as I later found out, had to be always on to prevent early filament melting. Added to this, while trying to unclog the nozzle on Sunday, I broke it clean in half, it seems to have been way too tight from the factory.
2. **Not the greatest stability:** The X-axis assembly, the one that holds the extruder, wasn't very stable. Because of this, the whole extruder assembly was dangling forward meaning that even if the extruder did work, the plastic might not stick very well to the build plate. Thus preventing the printer from, well, printing.

Now, on to the success criteria:

Assessment Rubric:

Level of achievement	How to achieve this level
1	The criteria isn't met.
2	The criteria is almost met, but wasn't for any reason. (Ex. time and resource management)
3	The criteria is met, but barely.
4	The criteria is met to an acceptable degree.
5	The criteria is fully met and maybe even surpassed.

Success criteria:

Success Criteria	Level of achievement According to me	Level of achievement According to the Survey
Build is stable and sturdy	I'd give it a 3 since the build is mostly stable, it's just the X-axis assembly that isn't very stable nor sturdy.	Poll says that the build is both stable and sturdy to a successful degree, giving it a 5 .
Build is done in time	If you go into the details then this is technically a 5, since it only mentions the build. But if you interpret it as it was probably interpreted it's a 2 . This is because the printer wasn't fully done in time.	The poll came to an average of 3 . Meaning the criteria is met, but barely.
3D printer has decent print quality	Since the printer doesn't really print yet, this criteria will have to get a 1 .	Since the printer didn't print by the day of the presentation, I decided to take out these criteria from the form. This was because there was no point in asking what they would rate it if I already knew it would be a 1 .
3Dprinter can print in PLA without heated bed	Same here, since I haven't gotten the printer to print yet, this criteria will also get a 1 .	
Electronics is not a wire mess	In this case I'd have to give it a 2 since the criteria wasn't really met due to lack of time to design and print parts that would manage/organize the cables.	The form unilaterally voted 4 . This means that the criteria was met to an acceptable degree.

What went wrong:

In a sentence, many, and I mean many, things went wrong. Here is a list with the major fails, why they happened and, if it applies, how I fixed it:

1. **The fallen project:** As I said before, while working on the project the whole X-axis fell to the floor and broke 4 of the five 3D printed pieces in that assembly. This was quite the issue since the print would take about 15 hours to complete meaning that it wouldn't have been done in time if it wasn't for a school activity that we had that Saturday which allowed me to start the print on Friday and then pick up the parts on Saturday. And let's just say I was so worried the print would fail, I literally had nightmares of the print failing.
2. **The filament the school had didn't have very good layer adhesion:** Layer adhesion is the name used to describe the bond between the layers of a print. And it just so happens that that wasn't very good in the filament the school had, causing parts to be fragile, which didn't help when the X-axis fell.
3. **During presentation day, the four previously broken parts broke again:** This I would attribute to 3 factors:
 - a. The layer adhesion
 - b. I didn't have the proper tools when removing the support which may have caused unnecessary stress to the part.
 - c. I was dumb.

Basically what happened was this: one of the four parts had broken while assembling the 3D printer and disassembling the whole printer again to fix it would take too long so I thought "I'll just remove the broken part, this won't cause too much stress on the joints" Well, turns out it did, which means that the 4 pieces were now broken again. Thankfully I had asked my mother to buy me some Poxipol (a very strong glue for plastic sold where I live) and I used that to fix the parts. But not before getting a panic attack which I thankfully dealt with well.

What went right:

Although many things went wrong, there were also a lot of successes. I will now proceed to list some of them:

1. **Once all the electronics were bought, they worked perfectly together.** They were all compatible with each other meaning the assembly process was pretty easy. As well as that, the software I picked also worked perfectly, only needing minor changes like calibration and some other customization changes.
2. **Excluding the X and Z axis assemblies, the build is pretty sturdy:** If we ignore the X and Z axis assemblies, the overall assembly of the Y axis and the base of the printer is actually pretty good. It is able to be lifted from the base without causing any damage to the structure and can handle all the vibrations coming from the motors perfectly.

What would I change now that I have the knowledge I have now?

Well, I've actually started changing things already, for example:

1. **The whole X-axis has been re-designed.** What I did here was instead of using the two rods to hold the extruder as I did before, I switched to using a 2020 extrusion which is quite sturdier than before.

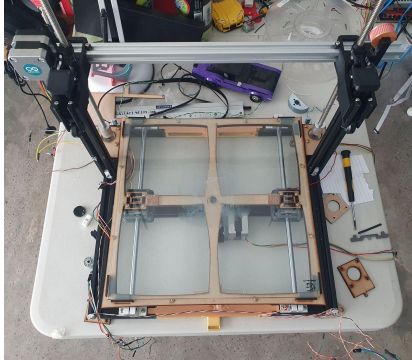


Fig 24: Riboldi, Matias. (17/3/2024). fig 24 [photo].

2. **The extruder mount has been made much sturdier:** Though it hasn't been designed yet at the time of writing this, the extruder mount will be designed to prevent the sagging that the previous one had. This will hopefully make the printer a lot less wobble-y and help to improve the print quality.
3. **The extruder has been mostly fixed:** The main issue, it seems, with the extruder was a clogged nozzle, well, and the fact that it was cut in half. Because of this I decided to buy a new nozzle, heat block and heat break. This means that I could just pretty much start from almost scratch again and actually do it properly this time.

As well as this, I have also realized, as you may have read in the research section, that the extruder I chose wasn't the best possible choice. While revising my research when I was writing this document, I realized that the BIQU H2 V2S would have been just as good, and maybe even better than the extruder I chose. All of this with really only one downside, the low availability of replacement parts for this extruder compared to the E3D one where I live.

All of these changes address issues such as stability, failure to extrude filament, and general sketchy-ness to make the 3D printer more reliable, and capable as a DIY printer.

Am I happy with how it went?

Meh, it's fine. The process was overall really fun to me, posing me with constant challenges to overcome. As well as that I have learned a lot which certainly makes me happy. Overall I'd say that I won't be at peace till I finish the project, and it won't be till then that I will truly be able to say if I liked how it went or not. But for now I'm quite happy that I don't have to make a report explaining what I did and didn't do with the new and upcoming parts of the 3D printer.

As to the report, I am mostly happy with how it looks and its hopefully not that hard to understand. I did try my best to explain the complicated parts in the simplest way possible for anyone to understand and enjoy reading this report.

Was the learning goal achieved?

I would say, it was mostly achieved. This is because I wasn't able to program the 3D printer myself due to time constraints. As well as that, the part of designing the electronics was quite ambitious since that, as well, would have required a lot more time. Everything else was achieved; I expanded my knowledge of programming and electronics to a new level and also learned a lot about 3D printing technology and how it works, what are its current limits and how to maybe, in the future, overcome them.

And to finish it off:

Here is a picture with all the failed/broken parts of the project.



Fig 25: Riboldi, Matias. (17/3/2024). fig 25 [photo].

All of these parts were either used to perfect the measurements of a part, as early prototypes of the printer, and if you look close enough you can see the broken hot end parts as well as some switches and a failed filament spool holder.

Sources:

E3D. (n.d.). *E3D Titan Extruder*. <https://e3d-online.com/products/titan-extruder>

3D Printer Parts - Page 1 - CRCibernética. (n.d.).
<https://www.crcibernetica.com/3d-printer-parts/>

E3D. (n.d.). *V6 All-Metal HotEnd*. <https://e3d-online.com/products/v6-all-metal-hotend>

Forman, V. a. P. B. E. (2019, November 26). *Stepper Motor*. helloCircuits!
<https://hellocircuits.com/2013/03/31/stepper-motor/>

Gharge, P. (2024, January 4). *The 5 best 3D printer Extruders in 2024*. Clever Creations. <https://clevercreations.org/best-3d-printer-extruder/>

Jbrazio. (2024, February 8). Boards. *Marlin Firmware*.
<https://marlinfw.org/docs/hardware/boards.html#:~:text=Marlin%20supports%20a%20wide%20variety.associating%20pins%20with%20their%20functions.>

Mahajan, P. (2023, December 28). 6 Best 3D Printer Extruder. *SelfCAD*.
<https://www.selfcad.com/blog/6-best-3d-printer-extruder>

Mensley, M. (2024, March 14). *The Best 3D Printers in 2024 – Buyer’s Guide*. All3DP.
<https://all3dp.com/1/best-3d-printer-reviews-top-3d-printers-home-3-d-printer-3d/>

RAMPS 1.4 - RepRaP. (n.d.). https://reprap.org/wiki/RAMPS_1.4

Sarf2k. (2024, February 8). Configuring Marlin. *Marlin Firmware*.
<https://marlinfw.org/docs/configuration/configuration.html>

Simplify3D Software. (2019, June 27). *Ultimate Materials Guide - Tips for 3D Printing with PLA*.

<https://www.simplify3d.com/resources/materials-guide/pla/#:~:text=Polylactic%20Acid%2C%20commonly%20known%20as,not%20require%20a%20heated%20b>

ed. Xometry, T. (2024, January 3). 5 Best 3D Printer Extruders. *Xometry*.
<https://www.xometry.com/resources/3d-printing/best-3d-printer-extruders/>

3D Printer Parts - Page 1 - CRCibernética. (n.d.).
<https://www.crcibernetica.com/3d-printer-parts/>

E3D. (n.d.). V6 All-Metal HotEnd. <https://e3d-online.com/products/v6-all-metal-hotend>

Forman, V. a. P. B. E. (2019, November 26). Stepper Motor. helloCircuits!
<https://hellocircuits.com/2013/03/31/stepper-motor/>

Gharge, P. (2024, January 4). The 5 best 3D printer Extruders in 2024. Clever Creations. <https://clevercreations.org/best-3d-printer-extruder/>

Jbrazio. (2024, February 8). Boards. Marlin Firmware.
<https://marlinfw.org/docs/hardware/boards.html#:~:text=Marlin%20supports%20a%20wide%20variety,associating%20pins%20with%20their%20functions.>

Mahajan, P. (2023, December 28). 6 Best 3D Printer Extruder. SelfCAD.
<https://www.selfcad.com/blog/6-best-3d-printer-extruder>

Mensley, M. (2024, March 14). The Best 3D Printers in 2024 – Buyer's Guide. All3DP.
<https://all3dp.com/1/best-3d-printer-reviews-top-3d-printers-home-3-d-printer-3d/>

RAMPS 1.4 - RepRap. (n.d.). https://reprap.org/wiki/RAMPS_1.4

Sarf2k. (2024, February 8). Configuring Marlin. Marlin Firmware.

<https://marlinfw.org/docs/configuration/configuration.html>

Simplify3D Software. (2019, June 27). Ultimate Materials Guide - Tips for 3D Printing with PLA.

<https://www.simplify3d.com/resources/materials-guide/pla/#::~:~:text=Polylactic%20Acid%2C%20commonly%20known%20as,not%20require%20a%20heated%20b>

ed. Xometry, T. (2024, January 3). 5 Best 3D Printer Extruders. Xometry.

<https://www.xometry.com/resources/3d-printing/best-3d-printer-extruders/>